INCH-POUND

MIL-PRF-1/49F
2 November 2007
SUPERSEDING
MIL-E-1/49E
27 April 1972

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6AS7G

Inactive for new design
after 21 July 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this specification sheet and MIL-PRF-1.

DESCRIPTION: High current, low Mu, double triode.

Outline --- 4-9.

Base --- B8-11.

Envelope --- ST16.

Cathode --- Coated unipotential.

Base connections:

Pin No. ---	1	2	3	4	5	6	7	8
Element ---	2g	2a	2k	1g	1a	1k	h	h

ABSOLUTE RATINGS:

Parameter:	Ef	Eb	Ec	Ib/p	Pp/p	Ehk	Rk/k	TE	Alt
Unit:	V	V dc	V dc	mA dc	W	v	Ohms	°C	ft
Maximum:	6.9	275	---	125	14.0	330	---	200	10,000
Minimum:	5.7	---	---	---	---	---	---	---	---
Test conditions:	6.3	135	0	---	---	---	250	---	---

GENERAL:

First article inspection: Required.

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TABLE I. Requirements or tests.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance Level (see note 6)	Symbol	Limits		Unit
					Min	Max	
	First article inspection						
1216	Base material insulating quality		---	---	---	---	---
	Conformance inspection, part 1						
1256	Electrode current (1) (anode)	See note 3	0.65	Ib	100	150	mA dc
1256	Electrode current (2) (anode)	Ec = -200 V dc; Eb = 250 V dc See note 3	0.65	Ib	---	10	mA dc
1266	Total grid current	Rg = 1.0 Meg Ω See notes 2 and 4	0.65	Ic	0	-4.0	µA dc
1231	Emission current	Eb = Ec = 10 V dc See note 5	0.65	Is	75	---	mA dc
1201	Shorts and discontinuity detection		0.4	---	---	---	---
	Conformance inspection, part 2						
1301	Heater current		6.5	If	2.26	2.74	A
1336	Heater-cathode leakage	See note 1	6.5	Ihk	---	50	µA dc
1031	Low-frequency vibration	Rp = 2,000 ohms; Ec = -7 V dc; Rk = 0 See note 1	6.5	Ep	---	200	mV ac
1306	Transconductance	See note 3	6.5	Sm	5,800	8,200	µmhos
1316	Amplification factor	See note 3	6.5	µ	1.4	2.6	---
1105	Permanence of marking		---	---	---	---	---
1101	Secureness of base, base insert, and cap		6.5	---	---	---	---
1111	Base pin solder depth		6.5	---	---	---	---

See footnotes at end of table.

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TABLE I. Requirements or tests - Continued.

MIL-STD-1311 method	Requirement or test	Conditions	Acceptance Level (see note 6)	Symbol	Limits		Unit
					Min	Max	
	Conformance inspection, part 3						
1501	Intermittent life-test (500 hours)	Group A; Rk = 125; Rg = 1 Meg Ω; Ehk = -300 V dc See note 1	---	---	---	---	---
---	Intermittent life-test end points (500 hours)						
1306	Transconductance		---	Sm	4,900	---	µmhos
1266	Total grid current		---	Ic	---	-10	µA dc
1336	Heater-cathode		---	Ihk	---	100	µA dc

NOTES:

1. Tie 1a to 2a; 1g to 2g; 1k to 2k.
2. With both units operating, Ic is the sum of I1c and I2c.
3. With both units operating, read each unit separately.
4. This test is to be performed at the conclusion of the holding period.
5. Read each unit separately: ground elements of opposite side; Rk = 0.
6. This specification sheet utilizes an accept on zero defect sampling plan in accordance with MIL-PRF-1, table III.

Referenced documents. In addition to MIL-PRF-1, this document references MIL-STD-1311.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5960-2007-027)

Review activities:
Army - AR
Navy - AS, CG, MC
Air Force - 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://www.assist.daps.dla.mil.

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